

$$\begin{aligned}
I(-2, -2) &= q^3 + q^3 \eta^{2v_0} + q^4 + q^4 \eta^{2v_0} + 2q^5 \\
I(-2, -1) &= \eta^{-v_0} (q^3 \eta^{v_0} + q^4 \eta^{-v_0} + 2q^4 \eta^{v_0} + q^5 \eta^{-v_0} + 2q^5 \eta^{v_0}) \\
I(-2, 0) &= \eta^{-2v_0} (q^3 + 2q^4 + q^5 \eta^{-2v_0} + 3q^5 + q^5 \eta^{2v_0}) \\
I(-2, 1) &= \eta^{-3v_0} (q^3 \eta^{-v_0} + 2q^4 \eta^{-v_0} + q^4 \eta^{v_0} + 2q^5 \eta^{-v_0} + q^5 \eta^{v_0}) \\
I(-2, 2) &= \eta^{-4v_0} (q^3 \eta^{-2v_0} + q^3 + q^4 \eta^{-2v_0} + q^4 + 2q^5) \\
I(-1, -2) &= \eta^{v_0} (-q^{1/2} \eta^{v_0} + q^{5/2} \eta^{v_0} + q^{7/2} \eta^{-v_0} + 3q^{7/2} \eta^{v_0} + 2q^{9/2} \eta^{-v_0} + 5q^{9/2} \eta^{v_0}) \\
I(-1, -1) &= -q - q^2 + q^3 + q^3 \eta^{2v_0} + q^4 \eta^{-2v_0} + 4q^4 + 2q^4 \eta^{2v_0} + q^5 \eta^{-2v_0} + 8q^5 + 2q^5 \eta^{2v_0} \\
I(-1, 0) &= \eta^{-v_0} (-q^{3/2} \eta^{-v_0} - q^{3/2} \eta^{v_0} + 2q^{7/2} \eta^{-v_0} + 2q^{7/2} \eta^{v_0} + 5q^{9/2} \eta^{-v_0} + 5q^{9/2} \eta^{v_0}) \\
I(-1, 1) &= \eta^{-2v_0} (-q - q^2 + q^3 \eta^{-2v_0} + q^3 + 2q^4 \eta^{-2v_0} + 4q^4 + q^4 \eta^{2v_0} + 2q^5 \eta^{-2v_0} + 8q^5 + q^5 \eta^{2v_0}) \\
I(-1, 2) &= \eta^{-3v_0} (-q^{1/2} \eta^{-v_0} + q^{5/2} \eta^{-v_0} + 3q^{7/2} \eta^{-v_0} + q^{7/2} \eta^{v_0} + 5q^{9/2} \eta^{-v_0} + 2q^{9/2} \eta^{v_0}) \\
I(0, -2) &= \eta^{2v_0} (-q - q^2 + q^3 \eta^{-2v_0} + q^3 + q^3 \eta^{2v_0} + q^4 \eta^{-2v_0} + 4q^4 + q^4 \eta^{2v_0} + 2q^5 \eta^{-2v_0} + 8q^5 + 2q^5 \eta^{2v_0}) \\
I(0, -1) &= \eta^{v_0} (-q \eta^{-v_0} - q \eta^{v_0} - q^2 \eta^{-v_0} - q^2 \eta^{v_0} + q^3 \eta^{-v_0} + q^3 \eta^{v_0} + 4q^4 \eta^{-v_0} + 4q^4 \eta^{v_0} + 8q^5 \eta^{-v_0} + 8q^5 \eta^{v_0}) \\
I(0, 0) &= 1 - 2q - 3q^2 + q^3 \eta^{-2v_0} + q^3 \eta^{2v_0} + 2q^4 \eta^{-2v_0} + 4q^4 + 2q^4 \eta^{2v_0} + 3q^5 \eta^{-2v_0} + 12q^5 + 3q^5 \eta^{2v_0} \\
I(0, 1) &= \eta^{-v_0} (-q \eta^{-v_0} - q \eta^{v_0} - q^2 \eta^{-v_0} - q^2 \eta^{v_0} + q^3 \eta^{-v_0} + q^3 \eta^{v_0} + 4q^4 \eta^{-v_0} + 4q^4 \eta^{v_0} + 8q^5 \eta^{-v_0} + 8q^5 \eta^{v_0}) \\
I(0, 2) &= \eta^{-2v_0} (-q - q^2 + q^3 \eta^{-2v_0} + q^3 + q^3 \eta^{2v_0} + q^4 \eta^{-2v_0} + 4q^4 + q^4 \eta^{2v_0} + 2q^5 \eta^{-2v_0} + 8q^5 + 2q^5 \eta^{2v_0}) \\
I(+1, -2) &= \eta^{3v_0} (-q^{1/2} \eta^{-v_0} + q^{5/2} \eta^{-v_0} + 3q^{7/2} \eta^{-v_0} + q^{7/2} \eta^{v_0} + 5q^{9/2} \eta^{-v_0} + 2q^{9/2} \eta^{v_0}) \\
I(+1, -1) &= \eta^{2v_0} (-q - q^2 + q^3 \eta^{-2v_0} + q^3 + 2q^4 \eta^{-2v_0} + 4q^4 + q^4 \eta^{2v_0} + 2q^5 \eta^{-2v_0} + 8q^5 + q^5 \eta^{2v_0}) \\
I(+1, 0) &= \eta^{v_0} (-q^{3/2} \eta^{-v_0} - q^{3/2} \eta^{v_0} + 2q^{7/2} \eta^{-v_0} + 2q^{7/2} \eta^{v_0} + 5q^{9/2} \eta^{-v_0} + 5q^{9/2} \eta^{v_0}) \\
I(+1, 1) &= -q - q^2 + q^3 + q^3 \eta^{2v_0} + q^4 \eta^{-2v_0} + 4q^4 + 2q^4 \eta^{2v_0} + q^5 \eta^{-2v_0} + 8q^5 + 2q^5 \eta^{2v_0} \\
I(+1, 2) &= \eta^{-v_0} (-q^{1/2} \eta^{v_0} + q^{5/2} \eta^{v_0} + q^{7/2} \eta^{-v_0} + 3q^{7/2} \eta^{v_0} + 2q^{9/2} \eta^{-v_0} + 5q^{9/2} \eta^{v_0}) \\
I(+2, -2) &= \eta^{4v_0} (q^3 \eta^{-2v_0} + q^3 + q^4 \eta^{-2v_0} + q^4 + 2q^5) \\
I(+2, -1) &= \eta^{3v_0} (q^3 \eta^{-v_0} + 2q^4 \eta^{-v_0} + q^4 \eta^{v_0} + 2q^5 \eta^{-v_0} + q^5 \eta^{v_0}) \\
I(+2, 0) &= \eta^{2v_0} (q^3 + 2q^4 + q^5 \eta^{-2v_0} + 3q^5 + q^5 \eta^{2v_0}) \\
I(+2, 1) &= \eta^{v_0} (q^3 \eta^{v_0} + q^4 \eta^{-v_0} + 2q^4 \eta^{v_0} + q^5 \eta^{-v_0} + 2q^5 \eta^{v_0}) \\
I(+2, 2) &= q^3 + q^3 \eta^{2v_0} + q^4 + q^4 \eta^{2v_0} + 2q^5
\end{aligned}$$